

Explainable Intent Fulfillment for Future Network Automation: Knowledge-Augmented Graph Guided Agent-to-Agent Collaboration

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Abstract—Intent-driven networking (IDN) is emerging as a key paradigm for future 6G network automation, yet translating flexible human intents into executable and trustworthy network service requests (NSRs) remains challenging. While large models offer strong capabilities for intent understanding, their probabilistic nature limits the explainability and trustworthiness required for carrier-grade telecom control. To address this issue, this paper proposes an explainable intent fulfillment framework that integrates agent-to-agent (A2A) collaboration with a telecom-grade Knowledge-Augmented Graph (KAG). Specifically, the framework coordinates three KAG-coupled agents: a semantic canonicalization agent that normalizes free-form intents into structured intent descriptors, a translation agent that generates candidate NSRs, and a KAG-grounded auditing agent that performs evidence retrieval, deterministic admission checking, and evidence-linked decision reporting. In an industrial URLLC slice-isolation prototype, the proposed framework improves NSR translation accuracy by 10% and strengthens the evidence-grounded traceability of edge-side decisions. These results demonstrate the potential of KAG-guided multi-agent collaboration for trustworthy and explainable 6G network automation.

Index Terms—6G network automation, intent-driven networking, explainable intent fulfillment, knowledge-augmented graph, multi-agent collaboration, large models.

I. INTRODUCTION

Intent-driven networking (IDN) is emerging as a key paradigm for future 6G network automation. This evolution is reflected in ETSI's Zero-touch Network and Service Management (ZSM) framework and 3GPP intent-driven management specifications, both of which elevate management abstractions beyond device-level commands and promote end-to-end automation for complex network services [1], [2], [3]. In parallel, recent advances in large models have created new opportunities for interpreting flexible human intents and supporting autonomous network decision-making. As a result,

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operators and vertical users can increasingly express service requirements in natural language, including latency guarantees, traffic isolation, reliability targets, and region-specific service priorities [4].

However, large models alone are insufficient for carrier-grade network control. Although LLMs are effective at interpreting flexible language and generating candidate configuration plans, their outputs remain probabilistic and may violate topology constraints, policy rules, or resource budgets. For mission-critical services, a generated decision is acceptable only when it is feasible, policy-compliant, reproducible, and explainable. This challenge becomes even more pronounced in multi-domain environments, where a single intent may simultaneously involve radio, transport, core, and edge/cloud resources [5]. Consequently, translating free-form intents into executable and trustworthy NSRs remains a fundamental challenge for future automated networks.

To address this challenge, we reformulate intent handling as *explainable intent fulfillment*. Rather than treating an intent as a prompt that directly triggers a generated network action, we cast fulfillment as a structured reasoning workflow in which free-form intent semantics are normalized, translated into candidate NSRs, and audited against explicit operational knowledge before execution. The proposed framework integrates Agent-to-Agent (A2A) collaboration with a telecom-grade Knowledge-Augmented Graph (KAG) [6]. Specifically, the framework coordinates three KAG-coupled agents: a semantic canonicalization agent that converts free-form intents into structured intent descriptors, a translation agent that generates candidate NSRs, and a KAG-grounded auditing agent that performs evidence retrieval, deterministic admission checking, and evidence-linked decision reporting. By coupling the semantic flexibility of large-model-based intent understanding with explicit knowledge grounding and verifiable control logic, the framework enables more trustworthy and explainable network automation.

The main contributions of this paper are summarized as follows. First, we propose an explainable intent fulfillment framework for future 6G network automation by integrating A2A collaboration with a telecom-grade KAG, in which three

specialized agents are responsible for semantic canonicalization, NSR translation, and KAG-grounded auditing, respectively. Second, we formulate intent fulfillment as a constrained and explainable decision workflow that maps free-form intents and local operational knowledge to executable NSRs through evidence retrieval, deterministic admission checking, and evidence-linked decision reporting. Third, we develop a prototype in an industrial URLLC slice-isolation scenario, and experimental results show that the proposed framework improves NSR translation accuracy by 10% while strengthening the evidence-grounded traceability of edge-side decisions.

II. BACKGROUND AND RELATED WORK

A. Agentic Intelligence for Network Automation

Recent advances in agentic intelligence have shifted AI systems from single-shot content generation to structured reasoning with tool use, task decomposition, and multi-agent coordination [7], [8]. This evolution is particularly relevant to network automation, where intent handling is inherently a multi-stage process involving intent understanding, semantic normalization, planning, verification, execution, and feedback-driven refinement. In this context, LLMs provide strong capabilities for interpreting telecom-domain language and generating candidate operational decisions, while agent-based collaboration improves modularity and scalability by assigning different reasoning and execution roles to specialized components.

Despite these advantages, existing agent-centric approaches are often optimized for semantic flexibility rather than carrier-grade operational reliability. In practical mobile and future network environments, an automatically generated action cannot be executed solely on the basis of linguistic plausibility. Instead, it must satisfy a series of hard constraints, including network topology dependencies, resource budgets, policy rules, security requirements, and Service-Level Agreement (SLA) targets [9]. As a result, purely agent-driven reasoning remains insufficient for dependable network automation unless it is grounded in explicit domain knowledge and complemented by deterministic verification mechanisms. This limitation motivates the integration of agentic reasoning with structured knowledge and auditable control logic.

B. Intent-Driven Networking and Explainable Fulfillment

Intent-Driven Networking (IDN) and Intent-Based Networking (IBN) aim to bridge the gap between high-level service objectives and low-level network configurations by exposing user or operator goals in an abstract form and translating them into executable control actions [10], [11]. Existing standards and prior studies have established the basic management pipeline of intent exposure, translation, realization, and assurance, thereby providing an important abstraction framework for network automation. However, many current implementations still rely heavily on template-based orchestration, predefined workflows, or heuristic control rules. Such approaches are often effective only in relatively stable scenarios, but they become increasingly brittle when intents are cross-domain,

context-dependent, semantically ambiguous, or dynamically evolving.

At the same time, explainability has emerged as a critical requirement for trustworthy network automation. In operational settings, the key challenge is not only whether an intent can be translated into an executable action, but also whether the resulting decision can be justified through inspectable evidence, validated against explicit constraints, and traced to observable execution outcomes [12], [13]. Existing IBN research provides useful abstractions for lifecycle management and service orchestration, whereas agentic AI research contributes flexible semantic reasoning capabilities [14]. Nevertheless, an important gap remains between these two lines of work: current solutions still lack an integrated mechanism that jointly supports flexible intent interpretation, explicit knowledge grounding, deterministic verification, and evidence-linked auditing. This gap directly motivates the framework proposed in this paper.

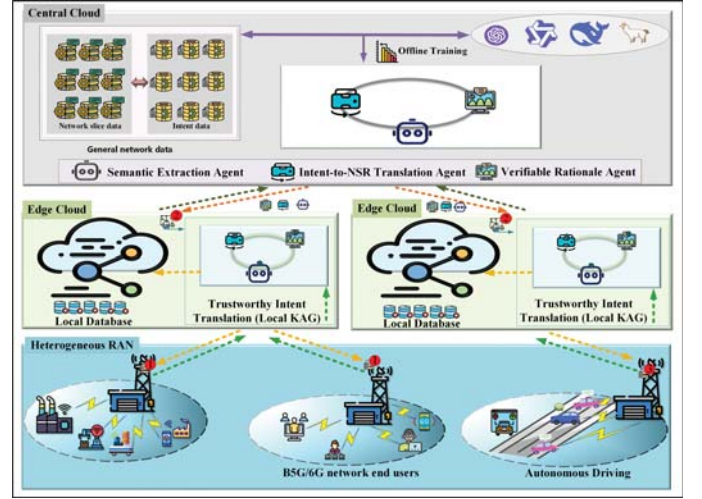


Fig. 1: Future network architecture for explainable intent fulfillment. The cloud governance plane maintains ontology, models, and policies; the edge plane performs intent canonicalization, A2A planning, KAG-based verification, and NSR generation.

III. SYSTEM ARCHITECTURE AND PROBLEM DESCRIPTION

A. System Architecture

As illustrated in Fig. 1, the proposed framework adopts a three-layer architecture consisting of the *Central Cloud*, multiple *Edge Clouds*, and the *Heterogeneous RAN*. The Central Cloud performs offline training and global governance by coordinating the semantic canonicalization module, the Intent-to-NSR Translation Agent, and the Rationale Agent, while maintaining globally shared knowledge artifacts for edge deployment. Each Edge Cloud conducts localized intent fulfillment by constructing a local KAG from site-specific topology, resource, policy, and telemetry data, and then executing online canonicalization, evidence retrieval, candidate NSR

generation, admission checking, and decision reporting. The Heterogeneous RAN serves as the execution layer, enforcing admitted NSRs and returning runtime feedback to the edge for closed-loop refinement. Overall, this hierarchical design separates global governance, local trustworthy translation, and network execution, thereby improving responsiveness while preserving feasibility, compliance, and explainability.

B. Problem Description

For notational brevity, the edge-site index is omitted in this subsection when no ambiguity arises. At time t , an edge site receives a raw natural-language intent x_t , which may specify service objectives such as latency, reliability, traffic isolation, or scenario-dependent resource preferences. Since such intents are not directly executable, the system first converts them into a canonical, machine-actionable representation

$$I_t = A_{\text{can}}(x_t) = \{s_t, q_t, k_t, p_t, h_t\}, \quad (1)$$

where s_t denotes the service type, q_t denotes the service scope, k_t denotes the KPI targets, p_t denotes priority and policy-related attributes, and h_t denotes time-horizon or contextual information.

The local operational database B_t is organized into a KAG snapshot

$$G_t = \Gamma(B_t) = (V_t, E_t, X_t), \quad (2)$$

where V_t and E_t denote the entities and relations in the local operational graph, and X_t stores node/edge attributes and runtime states such as topology availability, slice conditions, policy constraints, and telemetry measurements.

To support trustworthy decision making, the edge retrieves an intent-relevant evidence subgraph

$$\tilde{G}_t = R(I_t, G_t), \quad (3)$$

which contains the topology facts, capability constraints, slice states, policy clauses, and telemetry items most relevant to the current intent.

Conditioned on I_t and $\tilde{G}_t$, the translation agent generates a candidate NSR

$$\hat{n}_t = A_{\text{tr}}(I_t, \tilde{G}_t). \quad (4)$$

The candidate NSR is then verified through deterministic admission checking over the retrieved evidence subgraph to determine whether it can be safely dispatched under local topology, resource, SLA, and policy constraints. The final edge-side decision includes an action status, the admitted NSR when applicable, and an evidence-linked decision report.

In essence, the task is to map the pair (x_t, B_t) to an executable and explainable network decision under feasibility, compliance, and low-latency requirements. Only when the candidate NSR is admitted is it enforced by the RAN.

IV. PROPOSED METHOD

To align with the decision chain in Section III-B, the proposed method is organized into four stages: cloud-side supervised training and global governance, cloud-to-edge knowledge distillation, edge-side lightweight inference, and deterministic admission with closed-loop update.

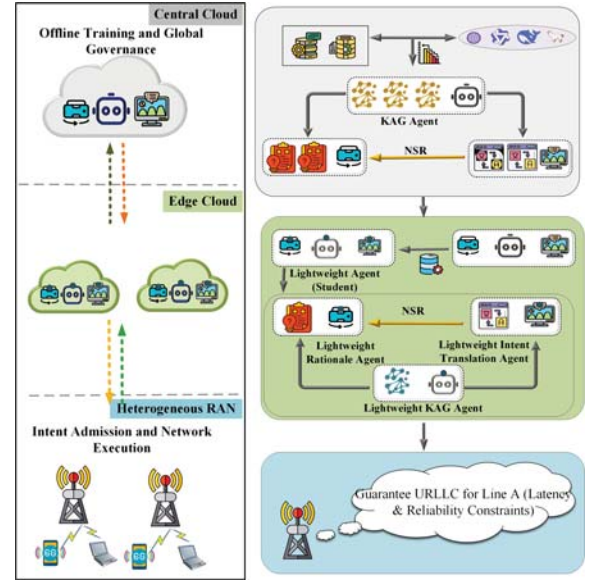


Fig. 2: Cloud-edge-RAN architecture for explainable intent fulfillment. The central cloud performs training, governance, and distillation, the edge cloud supports lightweight KAG-guided intent translation and verification, and the RAN executes admitted actions with runtime feedback for closed-loop refinement.

A. Cloud-Side Full-Agent Fine-Tuning and Global Governance

Let

$$\mathcal{D} = \{(x_i, G_i, n_i^{\text{ref}}, r_i^{\text{ref}})\}_{i=1}^N \quad (5)$$

denote the cloud-side training set, where x_i is the raw intent, G_i is the corresponding KAG snapshot, n_i^{ref} is the expert-verified reference NSR, and r_i^{ref} is the corresponding evidence-linked decision report.

During training, each raw intent is first canonicalized as in (1), and the cloud KAG agent retrieves an intent-relevant evidence subgraph

$$\tilde{G}_i = R(I_i, G_i), \quad (6)$$

which contains the topology facts, slice states, capability constraints, policy clauses, and telemetry items most relevant to the current intent.

On top of the retrieved evidence, the central cloud hosts two full-capability teacher agents: a translation teacher A_{tr}^T and a rationale teacher A_{ra}^T . The former maps a canonical intent to a reference NSR, whereas the latter generates the corresponding evidence-linked decision report. Their supervised objective is

$$\begin{aligned} \mathcal{L}_T = & - \sum_{i=1}^N \log p_{\theta_{\text{tr}}^T} \left(n_i^{\text{ref}} \mid I_i, \tilde{G}_i \right) \\ & - \lambda \sum_{i=1}^N \log p_{\theta_{\text{ra}}^T} \left(r_i^{\text{ref}} \mid I_i, n_i^{\text{ref}}, \tilde{G}_i \right), \end{aligned} \quad (7)$$

where θ_T^{tr} and θ_T^{ra} denote the parameters of the two teacher agents, and λ balances NSR generation and decision-report generation.

In parallel, the cloud governance plane maintains the global ontology, policy bundle, and agent registry. These globally consistent artifacts ensure that all edge sites operate under the same semantic schema, verification rules, and service abstractions.

B. Cloud-to-Edge Knowledge Distillation

Although the cloud teachers provide strong semantic and reasoning capability, directly deploying them to the edge is impractical under latency and hardware constraints. We therefore distill the cloud teachers into a lightweight student backbone together with two task-specific student heads: a translation student A_{tr}^S and a rationale student A_{ra}^S .

Let $p_{T,\text{tr}}^\tau$ and $p_{S,\text{tr}}^\tau$ denote the softened token distributions of the translation teacher and student, respectively, and let $p_{T,\text{ra}}^\tau$ and $p_{S,\text{ra}}^\tau$ denote those of the rationale teacher and student. The distillation loss is defined as

$$\mathcal{L}_{\text{KD}} = \tau^2 \sum_{i=1}^N \text{KL}(p_{T,\text{tr}}^\tau \| p_{S,\text{tr}}^\tau) + \lambda \tau^2 \sum_{i=1}^N \text{KL}(p_{T,\text{ra}}^\tau \| p_{S,\text{ra}}^\tau), \quad (8)$$

where τ is the distillation temperature.

The student optimization objective is then

$$\begin{aligned} \mathcal{L}_S = & - \sum_{i=1}^N \log p_{\theta_S^{\text{tr}}}(n_i^{\text{ref}} | I_i, \tilde{G}_i) \\ & - \lambda \sum_{i=1}^N \log p_{\theta_S^{\text{ra}}}(r_i^{\text{ref}} | I_i, n_i^{\text{ref}}, \tilde{G}_i) \\ & + \mu \mathcal{L}_{\text{KD}}, \end{aligned} \quad (9)$$

where μ controls the strength of distillation.

After distillation, the cloud packages the student parameters together with the ontology, policy bundle, and site-specific governance artifacts, and then distributes them to the edge cloud for low-latency execution.

C. Edge-Side Lightweight KAG-Guided Inference

At runtime, an edge site receives a raw intent x_t and first canonicalizes it into

$$I_t = A_{\text{can}}^S(x_t) = \{s_t, q_t, k_t, p_t, h_t\}, \quad (10)$$

where s_t denotes the service type, q_t denotes the service scope, k_t denotes the KPI targets, p_t denotes priority and policy-related attributes, and h_t denotes the time-horizon or contextual information.

The lightweight KAG agent then retrieves the local evidence subgraph

$$\tilde{G}_t = R_S(I_t, G_t), \quad (11)$$

which serves as the operational context for edge-side reasoning.

Conditioned on I_t and $\tilde{G}_t$, the lightweight translation agent generates a candidate NSR

$$\hat{n}_t = A_{\text{tr}}^S(I_t, \tilde{G}_t). \quad (12)$$

This design matches the cloud–edge split in Fig. 2, the cloud provides high-capacity semantic training and governance, whereas the edge executes low-latency candidate generation on top of local KAG evidence.

D. Deterministic Admission, Decision Reporting, and Closed-Loop Refinement

Before dispatch, the candidate NSR must pass deterministic admission against the retrieved KAG evidence. We denote the verification result by

$$z_t = C(\hat{n}_t, \tilde{G}_t), \quad z_t \in \{0, 1\}, \quad (13)$$

where $C(\cdot)$ checks topology feasibility, resource availability, SLA consistency, and policy compliance.

After verification, the lightweight rationale agent generates an evidence-linked decision report

$$r_t = A_{\text{ra}}^S(I_t, \hat{n}_t, \tilde{G}_t). \quad (14)$$

The final edge-side output is therefore

$$(a_t, n_t^{\text{adm}}, r_t) = \begin{cases} (\text{admit}, \hat{n}_t, r_t), & z_t = 1, \\ (\text{refine/reject}, \emptyset, r_t), & z_t = 0. \end{cases} \quad (15)$$

If the candidate is admitted, the network execution plane enforces n_t^{adm} and returns runtime observations y_t , including logs, KPI measurements, alarms, and violation events. The local KAG is then updated as

$$G_{t+1} = \Psi(G_t, y_t). \quad (16)$$

In this way, the system forms an explainable closed loop in which only admitted NSRs are dispatched, while every edge-side decision remains linked to the evidence used during generation and verification.

V. EXPERIMENTS

We evaluate the proposed framework in an industrial URLLC isolation scenario, where the system processes intents such as “isolate Line A control traffic with strict latency and reliability guarantees.” As shown in Fig. 2, the central cloud handles offline training and global governance, while the edge cloud performs online KAG-guided candidate generation, deterministic admission, and evidence-linked decision reporting. All methods are tested on the same intent set against expert-verified reference NSRs.

This prototype focuses on two directly observable outputs of the edge decision loop: the structural correctness of generated NSRs and the grounding quality of evidence in edge-side decision reports, while cloud–edge latency, model compression ratio, and long-horizon closed-loop convergence are left to an extended version. For NSR generation, we compare KAGLlama, Llama, and Transformer. For evidence-grounding

evaluation, only KAGLlama (Edge) and Llama (Edge) are included.

To evaluate NSR generation quality, we use the Levenshtein accuracy between the generated NSR $\hat{n}_i$ and the expert-verified reference NSR n_i^{ref} :

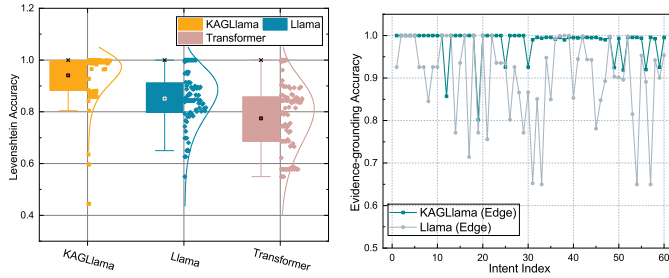
$$\text{Acc}_{\text{Lev}}(\hat{n}_i, n_i^{\text{ref}}) = 1 - \frac{d_{\text{Lev}}(\hat{n}_i, n_i^{\text{ref}})}{\max(|\hat{n}_i|, |n_i^{\text{ref}}|)}, \quad (17)$$

where $d_{\text{Lev}}(\cdot, \cdot)$ denotes the Levenshtein edit distance.

To evaluate evidence grounding at the edge, let $\hat{E}_i = \{e_{i,1}, \dots, e_{i,K_i}\}$ denote the evidence items attached to the generated decision report. We define

$$\text{Acc}_{\text{Evd}}(i) = \frac{1}{K_i} \sum_{k=1}^{K_i} \mathbf{1}(e_{i,k} \in \tilde{G}_i \wedge \text{Supports}(e_{i,k}, \hat{n}_i)), \quad (18)$$

where $\mathbf{1}(\cdot)$ is the indicator function, and $\text{Supports}(e_{i,k}, \hat{n}_i)$ indicates that the cited evidence item is consistent with at least one field or constraint appearing in the generated NSR.



(a) NSR generation accuracy under URLLC isolation. (b) Evidence-grounding accuracy across intents.

Fig. 3: Edge-side performance comparison of intent translation results.

Fig. 3a shows the Levenshtein accuracy distributions under the URLLC isolation scenario. KAGLlama achieves the highest accuracy among the three methods, indicating that KAG-guided translation produces NSRs that are more consistent with expert-verified references. Llama shows lower and more dispersed accuracy, while Transformer performs worst overall, suggesting that explicit knowledge grounding is important for structurally constrained NSR generation.

Fig. 3b reports the evidence-grounding accuracy across edge intents. KAGLlama (Edge) remains close to 1.0 for most intent instances, whereas Llama (Edge) exhibits larger fluctuations and several obvious drops. This result indicates that KAG grounding provides more reliable evidence support for edge-side decision reports.

Overall, the current prototype verifies the benefit of KAG grounding in two observable aspects: structural fidelity of generated NSRs and grounding quality of attached evidence. Since this conference version does not yet include dedicated cloud-edge latency or distillation benchmarks, our experimental claims are intentionally restricted to these two aspects.

A. Conclusion

This paper investigated explainable intent fulfillment for future network automation and proposed a KAG-guided cloud-edge-RAN collaborative framework. By combining large-model-based intent understanding with knowledge-grounded validation and evidence-linked execution, the framework improves feasibility, transparency, and auditability in intent-driven networking. The results suggest that future 6G automation should integrate semantic flexibility with explicit operational knowledge rather than relying solely on black-box generation. Future work will address multi-domain scalability, dynamic knowledge maintenance, and unified evaluation of automation quality and explainability.

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